(b) To calculate the resistance of the bulbs, we use the relation $P = \frac{V^2}{R}$. In both the bulbs, the voltage drop is the same. So the power is inversely proportional to the resistance or resistance is inversely proportional to the power $\left(R \propto \frac{1}{P}\right)$. It implies that, the 30W has twice as much as resistance as 60 W bulb.

(c) When the bulbs are connected in series, the current passing through each bulb is the same. It is equivalent to two resistors connected in series. The bulb which has higher resistance has higher voltage drop. So 30W bulb will glow brighter than 60W bulb. So the higher power rating does not always imply more brightness and it depends whether bulbs are connected in series or parallel.

EXAMPLE 2.16

Two electric bulbs marked 20 W – 220 V and 100 W – 220 V are connected in series to 440 V supply. Which bulb will get fused?

Solution

To check which bulb will get fused, the voltage drop across each bulb has to be calculated.

The resistance of the bulb,

$$R = \frac{V^2}{P} = \frac{(\text{Rated voltage})^2}{\text{Rated power}}$$

For 20W-220V bulb,

$$R_1 = \frac{(220)^2}{20}\Omega = 2420\,\Omega$$

DQNPP1

For 100W-220V bulb,

$$R_2 = \frac{(220)^2}{100}\Omega = 484\,\Omega$$

Both the bulbs are connected in series. So same current will pass through both the bulbs. The current that passes through the circuit, $I = \frac{V}{R_{tot}}$.

$$R_{tot} = (R_1 + R_2)$$

$$R_{tot} = (484 + 2420)\Omega = 2904\Omega$$

$$I = \frac{440V}{2904\Omega} \approx 0.151\text{A}$$

The voltage drop across the 20W bulb is

$$V_1 = IR_1 = \frac{440}{2904} \times 2420 \approx 366.6\text{ V}$$

The voltage drop across the 100W bulb is

$$V_2 = IR_2 = \frac{440}{2904} \times 484 \approx 73.3\text{ V}$$

The 20 W bulb will get fused because the voltage across it is more than the voltage rating.

2.4 ELECTRIC CELLS AND BATTERIES

An electric cell converts chemical energy into electrical energy to produce electricity. It contains two electrodes (carbon and zinc) immersed in an electrolyte (sulphuric acid) as shown in Figure 2.17.

Several electric cells connected together form a battery. When a cell or battery is connected to a circuit, electrons flow from the negative terminal to the positive terminal through the circuit. By using chemical reactions, a battery

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